SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics

NAME: Hariny A P **COURSE CODE:** CSA1506

REG NO: 192511334

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

Code of Conduct

I, Hariny A P / 192511334 (Name / Reg No), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Hariny A P (192511334)**

Assessment Tasks

Short Answer Test

SET 1

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of data centers in delivering cloud services.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

CO1-AT3 - SHORT ANSWER TEST

NAME: Harini A P

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COURSE CODE: CSA1506

COURSE NAME: CLOUD COMPUTING AND BIG DATA ANALYTICS.

Q1. CLOUD COMPUTING:

DEFINITION:

Cloud computing is defined as the delivery of computing resources such as servers, storage, networks and software over the internet based on user demands.

CHARACTERISTICS:

↳ On-Demand Self-Service: Users can acess computing resources whenever needed without contacting any cloud provider.
Eg: VM on AWS.

↳ Broad Network Access: Cloud services can be accessed over the internet using laptops, smartphones ..etc.
Eg: Google Drive.

↳ Resource Pooling: The provider shares computing resources among multiple customers while keeping their data separated.

↳ RAPID ELASTICITY: Resources can be increased or decreased according to user demand.

↳ MEASURED SERVICE: Resource usage is monitored and customers are charged according to their consumption.

Traditional computing requires organizations to purchase and maintain their own hardware, whereas cloud computing provides flexible resources without requiring users to own the underlying infrastructure.

Q2. EVOLUTION OF CLOUD COMPUTING:

Cloud computing evolved through several technological developments.

1. Mainframe Computing: Large centralised computers allow multiple users to share computing resources.
2. Personal computers and client-server computing: Personal computers became widely available while servers provide centralised access to data and services to clients.
3. Internet Development: The growth of the internet enabled users to access information and services from different locations.
4. Virtualization: A single physical server could run multiple virtual machines, improving resource utilization and reducing hardware costs.
5. Web-Based Software: Applications became accessible through browsers, leading to the growth of Software as a Service (SaaS)
6. Modern Cloud Platforms: Providers began offering computing powers, storage, databases and developmental platforms on demand

Eg: AWS, Google Cloud, Microsoft Azure.

Q3. SERVER VIRTUALIZATION: It is a technology that divides a single physical server into multiple independent virtual machines. A software layer called hypervisor manages these virtual machines.

Role of Server Virtualization:

1. Resource Utilization: Multiple virtual machines can share a physical server efficiently.
2. Cost Reduction: Fewer physical servers are needed, reducing hardware and maintenance expenses.
3. Isolation: Each VM can run its own operating system and applications independently.
4. Scalability: VM's can be created or removed according to demand and workload requirements.
5. Flexibility: VM's can often be moved between compatible physical hosts.

VIRTUALIZATION	CLOUD - COMPUTING.
★ Virtualization is an enabling technology.	★ It is a service delivery model.
★ Creates virtual version of physical resources	★ Provides computing resources as service.
★ Do not automatically provide pay-per-use billing	★ Commonly supports on-demand access and usage based billing.
★ Example: VMware ESXi	★ Example: Amazon EC2.

Q4. DATA CENTRES: It is a facility containing servers, storage systems networking equipments, power supplies and cooling systems used to process, store and manage data.

ROLE OF DATA CENTRES:

1. Computing: Servers execute applications, process requests and run virtual machines.
2. storage: storage systems maintains files, databases, backups and application data.
3. Networking: Routers and Switches connect servers and enables users to access cloud services over the internet.
4. Resource Management: Virtualizations and management software allocate resources efficiently among customers.
5. Reliability: Backup systems, redundant power supplies, and network connections help minimize service interruptions.
6. Security: physical access controls, surveillance, and technical security measures protect infrastructure and data.
7. Scalability: Additional computing and storage resources can be deployed to meet increasing demand.

Example: Google Drive stores user file in cloud infrastructure hosted in data centres and makes them accessible through internet.

Q5. Cloud service models differ according to the resources provided and the responsibilities handled by the user and provider.

1. Infrastrure as a Service (IaaS):

Provides virtual machines, storage and networking. Users manage the operating system and applications.

Eg: Amazon EC2.

2. Platform as a Service (PaaS):

Provides a platform for developing, testing and deploying applications without directly managing the underlying infrastructure.

Eg: Google App Engine.

3. Software as a Service:

Provides complete software applications over the internet, usually through a browser.

Eg: Google Workspace, such as Gmail and Docs.

4. Anything as a Service: (Xaas)

A broad term for delivering various IT capabilities including infrastructure, software, platforms and security as service.

Eg: Cloudflare's cloud-based security service.

MODEL	WHAT IT PROVIDES	EXAMPLE
IaaS	Computing infrastructure	Amazon EC2
PaaS	Application development platform	Google App Engine
SaaS	Ready to use software	Google Workspace
XaaS	Various IT services	cloudflare security services.